Code No: 136BC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is the need of design pattern?	[2]
b)	How would you select design patterns?	[3]
c)	How would you describe formatting document?	[2]
d)	Write about advantages of user interface.	[3]
e)	State about the prototype.	[2]
f)	How would you explain about builder for construct objects?	[3]
g)	Illustrate about the proxy.	[2]
h)	What is the use of adapter?	[3]
i)	Outline about the mediator.	[2]
j)	How would you summarize the interpreter?	[3]

PART – B

(50 Marks)

2.a)	Explain about organizing the catalog. And mention the importance of it.	
b)	Can you explain the how to solve the complex problems use of design patterns.	[5+5]
	OR	
3.a)	Summarize the catalog of design patterns.	
b)	Explain about design patterns in smalltalk MVC.	[5+5]
4.a)	Discuss about document structure and explain design problems.	
b)	Mention the importance of multiple look and feel standards.	[5+5]
	OR	
5.a)	List and explain user operations for spelling checking.	
b)	Summarization of design document patterns editor.	[5+5]
6.a)	Can you justify yours answer, how abstract factory create a relationship between groups.	
b)	Illustrate the factory methods.	[5+5]
	OR	
7.a)	Explain prototype patterns with suitable examples.	
b)	Discuss about singleton. And mention the need of it.	[5+5]

8.a) Demonstrate composite patterns composes objects as trees with suitable example .
b) Can you elaborate the proxy for structural pattern matching? [5+5]

OR

9.a) Examine the facade interface for structural patterns.
b) Can you explain how flyweight pattern saves the RAM? [5+5]

10.a) Demonstrate the template method with suitable example.
b) Explain the following:
i) State
ii) Observer. [5+5]

OR

11.a) Summarize the responsibility for behavioral patterns.
b) Explain about strategy pattern algorithms for specific task with suitable example. [5+5]

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